Evaluating Effects of Word Learning on Neural Entrainment to Speech

Background Electroencephalographic (EEG) recordings during speech listening show that low frequency (~3-8 Hz) components of the speech amplitude envelope and listeners' brain activity are phase correlated, a phenomenon known as neural entrainment. Neural entrainment increases with speech intelligibility and can distinguish two individual speaker's voices when mixed within the same recording. These types of findings suggest promise as a tool in neurophysiological research on real time, continuous speech comprehension. Such tools have have been largely unavailable until now, in contrast to the rich literature on ERPs elicited during sentence reading. However, the utility of neural entrainment is limited by a significant dearth of understanding about its underlying mechanisms and antecedent conditions. While some findings suggest that entrainment is closely tied to low-level sensory processing, including the fact that entrainment occurs even in the absence of attention, effects of intelligibility on entrainment could indicate that it reflects comprehension. If brain-speech entrainment is to be a productive tool for studying speech comprehension, we need clearer understanding of whether it reflects sensory processing, high-level language comprehension processes, or some mixture.

To investigate the mechanisms of brain-speech entrainment, we tested the hypothesis that entrainment during artificial language listening would increase as listeners acquire semantic knowledge about the words in the language. We predicted that neural entrainment during listening to a mostly unknown language would increase in moments after hearing a known word, compared to other moments. If entrainment mainly reflects sensory processing of the acoustic input, it should not be affected by word knowledge. The larger motivation for our work is to develop a task-free measure of word comprehension during spoken language learning, which can be used to track the impacts of learning in real time.

Design and Procedure During EEG recording participants listened to a 9 minute recording of pseudoword speech, which was derived from an English story by replacing words with pseudowords (**figure 1**) and narrated by a native English speaker. Next, in a word learning phase, participants learned the English translations of 10 pseudowords from the narrative, and were familiarized with 10 other pseudowords without learning their meanings (**figure 2**). Finally, after the word learning phase, participants heard the same pseudoword narrative a second time. Participants also listened to a 9-minute English narrative once, either before or after the pseudoword narratives (order was randomly assigned to participants). The narratives were based on excerpts from *The Wizard of Oz* and*The Velveteen Rabbit*. For each participant, the English narrative was one of these two excerpts, and the pseudoword narrative was derived from the other excerpt (counterbalanced across participants). Our main prediction was that entrainment during the second pseudoword story exposure would increase in moments following meaningful words, relative to moments following the merely familiarized words.

Analyses. Our analyses are still in progress, so we will not have results to report until a few days after the HSP submission deadline. We measured entrainment in two ways: 1) the phase coherence between the EEG and speech signals; and 2) the cross-correlation between the EEG and speech. We band-pass filtered (4-8 Hz) the EEG activity and the corresponding speech amplitude envelope and then extracted instantaneous phase values for each of these signals, using the Hilbert transform. We next calculated the difference in instantaneous phase values between the band-limited EEG and speech envelope at each time point in the epoch. We calculated phase coherence as the consistency of the phase difference between EEG and speech envelopes across the epoch, yielding a value ranging from 0 to 1. We calculated the cross-correlation between EEG and speech by sliding the two band-limited signals across each other at multiple lags and identifying the maximum inner product between them. Quantifying entrainment in these two ways, we tested our main hypothesis by asking whether entertainment during the second block of pseudoword speech was greater in moments after a newly learned word, compared to moments after a merely familiarized word. We also compared entrainment to English and pseudoword speech, predicting greater entrainment to English, and we compared entrainment in the first and second repetitions of the pseudoword speech.

1. **English story**

"Dorothy lived in the midst of the great Kansas prairies, with Uncle Henry, who was a farmer, and Aunt Em..."

2. English-pseudoword translation dictionary

English	Pseudoword
Dorothy	avinod
lived	klice
uncle	wotho
kansas	wobud
farmer	vatic
who	tos

3. Pseudoword translation of English story

"Avinod klice nuv sho riste zeth sho adeal wobud deceival, suv wotho lofa, tos rie e vactic, bol ponk ig..."

Figure 1. Stimulus generation procedure.

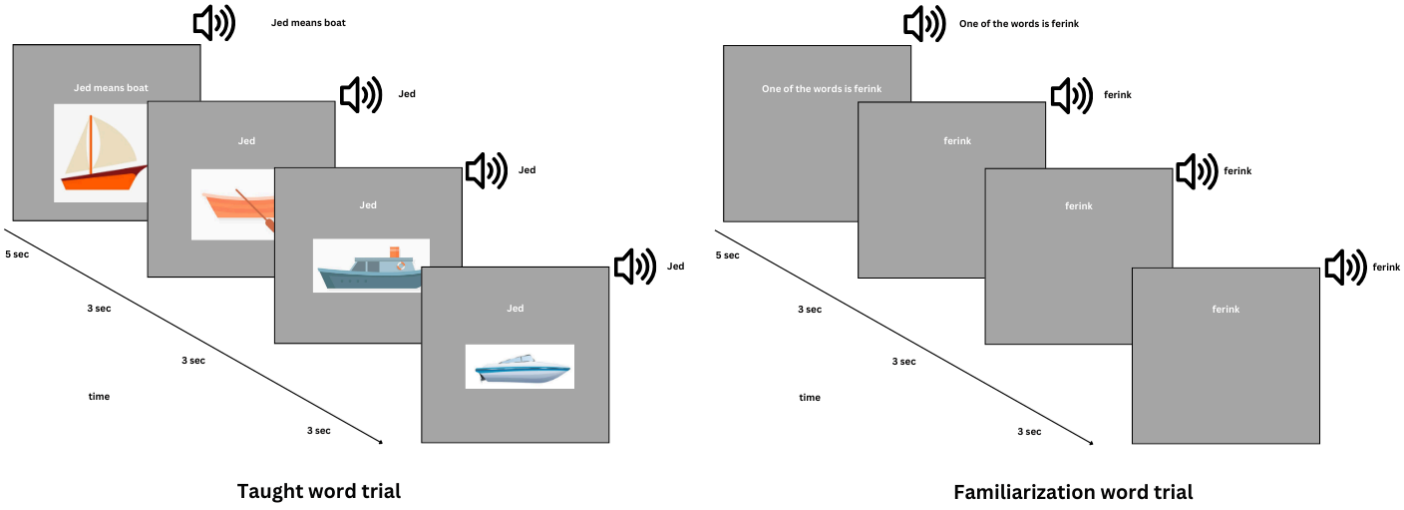

Figure 2. Example taught and familiarized word trial structures. For each of the 10 taught and 10 familiarized pseudowords, participants saw two such four-frame trials in a fixed random order.